

AI FOR DIGITAL PUBLIC SAFETY

Defeating Counterfeiting, Fraud & Digital Arrest Scams

Theme: Smart Cities / Public Safety / Digital Trust

Google Colab & CLI Demonstration Notebook Ready

Problem Context: The Crisis of Digital Trust

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1.14 Million Cybercrime Complaints (2023)

India registered a massive 60% surge in complaints compared to the previous year.

Digital Arrest Scams are Industrialised

Organised networks impersonating CBI, ED, TRAI, and Customs operate out of cross-border compounds.

Rs 1,776 Crore Scammed in 9 Months

MHA reports record financial losses from victims trapped in psychological hostage situations over video call.

Fake Indian Currency Notes (FICN) Threat

RBI's Annual Report 2025 flagged high-quality counterfeit Rs 500 bills bypassing manual verification.

The Challenge Statement

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Build an AI-powered Digital Public Safety Intelligence platform that equips law enforcement agencies, financial institutions, and citizens with proactive tools to detect, disrupt, and respond to digital fraud networks, counterfeit currency circulation, and organised scam operations — shifting from reactive case investigation to predictive threat neutralisation.

Why This Platform is Different

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Unlike standalone systems, our solution combines seven specialized AI agents that collaboratively analyze conversation content, voice biometrics, counterfeit currency, fraud graphs, and geospatial intelligence before generating an explainable, legally auditable recommendation.

This multi-agent architecture reduces false positives, improves transparency, and enables proactive interventions before financial loss occurs.

Solution Overview & Agentic Architecture

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Agentic AI Orchestrator

Acts as the central command, routing inputs to domain-specific agents and fusing their signals.

Five Specialized Subsystems

Scam classification, speech forensics, citizen chatbot shield, counterfeit CV check, graph network mapping.

Cryptographic Audit Ledger

Ensures all intelligence logs and findings are for ensic-grade and court-admissible.

Google Colab & CLI Support

Reconciled constraints to run seamlessly as a Python package, CLI utility, or Colab notebook.

Module 1: ScamShield Agent (Digital Arrest NLP)

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Multi-Engine Pattern Check

Supports three modes of analysis: rule Heuristic, machine learning, and advanced LLM.

Threat Progression Tracking

Identifies current stage (e.g. Isolation) and predicts next likely stage (Payment Demand).

Explainability Layer

Highlights specific triggered indicators (Fake Authority, Psychological Coercion, Block Threats).

Logistic Regression Rationale

Selected because it provides explainable probabilities suitable for forensic NLP and small-to-medium datasets.

Module 2: VoiceGuard Agent (Speech AI Forensics)

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Acoustic Feature Extraction

Simulates auditing of voice pitch variance, spectral roll-off, and centroid std.

Jitter and Shimmer Anomalies

Identifies synthetic voices by analyzing the absence of micro-modulations (Li et al. 2023).

Voice Cloned Flagging

Alerts telecom systems when incoming speech streams match machine-synthesized speech signatures.

Lightweight OpenCV Rationale

Allows lightweight feature processing and verification checks without requiring GPU inference.

Module 3: CitizenShield Agent (Conversational Chatbot)

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Conversational Evaluation

Provides instant risk verdicts for messages, links, and transaction requests.

Regional Advisory Support

Supplies guided prevention steps in 4 regional languages (English, Hindi, Telugu, Tamil).

NCRP-Style Incident Reports

Autogenerates structured cyber incident reports matching NCRP workflows for manual submission.

12 Language Scalability Goal

Designed to expand to 12 regional languages using multilingual LLMs or Translate APIs.

Module 4: NoteGuard Agent (Explainable CV Checksheet)

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Computer Vision Checksheet

Audits notes using explicit checks (Security Threat, Watermark, UV, Serial Prefix).

Random Forest Rationale

Selected because it performs well on structured handcrafted counterfeit features while remaining explainable and lightweight.

Watermark & Microprint Check

Evaluates sharpness using Laplacian variance and BGR channel distributions (PASS/FAIL).

Deep Learning Fallback CNN

Future work trains CNNs on rupee images to achieve >97% accuracy (Patil et al. 2022).

Module 5: FraudGraph Agent (Network Mule Rings)

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NetworkX Modeling Rationale

Enables interpretable graph analytics and rapid fraud network visualization for investigative workflows.

Syndicate Clustering

Traces transaction edges to discover coordinated rings utilizing shared device fingerprints.

Reveal Hidden Fraud Rings

Serves as a magnifying glass (Linkurious) exposing illicit links traditional systems miss.

Admissible Court Evidence Package

Generates comprehensive JSON dossiers compiling all linked victims and transactions.

Module 6: GeoWatch Agent (Geospatial Patrols)

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GIS Statistical Crime Mapping

Fuses crime rates, incident counts, and financial losses to generate risk rankings.

Incident Breakdown Reasons

Displays the specific threat factors (number of complaints, spoofed numbers, UPI links).

Actionable LEA Recommendations

Deploy cyber investigation units, recommend telecom operator notification, and push awareness SMS.

Interactive Leaflet HTML Map

Renders a dark-theme HTML map pinpointing district markers sized by financial damage.

Real-time Interception and Alerting

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Active Interception Webhooks

Simulates instant API notifications pushed to Jio, Airtel, and Vi telecom circles.

Fast Reaction Latency

Average detection-to-alert latency clocked at under 0.45 seconds.

Massive Lead Time Gained

Flags scam sessions 15-45 minutes before financial transfer occurs.

INR Losses Prevented

Estimated savings calculated across batch runs, preventing lakhs in fraud transfers.

Weighted Threat Risk Fusion Engine

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Multi-Modal Risk Correlator

Correlates distinct risks: Text (NLP), Voice (Acoustic), Graph (Mule), and Geo (Hotspot).

Weighted Threat Formula

Applies: $\text{FusedScore} = 0.4 * \text{Text} + 0.2 * \text{Voice} + 0.2 * \text{Graph} + 0.2 * \text{Geo}$.

Unified Threat Ratings

Translates the fused score into priority levels (CRITICAL, HIGH, MEDIUM, LOW).

Weight Selection Rationale

Text (40% primary intent), Voice (20% acoustics), Graph (20% mule links), Geo (20% context).

Human-in-the-Loop (HITL) Admissibility

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LEA Decision Support

The AI system does not issue automated arrests or bank blocks without verification.

Officer Review Dashboard

Enables officers to inspect auto-drafted court evidence packages via the CLI interface.

Cryptographic Sign-Off

Upon approval, the officer's choice is cryptographically signed and hashed in the ledger.

Tamper-Proof Ledger

Section 65B compliance is validated by ensuring that approved FIR drafts remain locked.

Responsible AI & Deployment Scope

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Decision Support System

Designed to assist investigators and financial institutions; final operational decisions remain with authorized personnel.

Synthetic Evaluation Data

Prototype models are evaluated using carefully generated synthetic datasets due to limited public fraud data availability.

Simulated Integrations

Telecom notifications, government alerts, and NCRP workflows are demonstrated as simulated integrations.

Officer Oversight Lock

All enforcement actions require officer review through the Human-in-the-Loop (HITL) workflow before escalation.

Future Technical Roadmap

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Voice Streaming Forensics

Real-time analysis of active phone/VoIP calls using low-latency streaming biometrics.

Edge AI Counterfeit Detection

Optimizing NoteGuard to execute Random Forest & CNN models directly on low-power mobile GPUs.

Graph Neural Networks (GNNs)

Enhancing FraudGraph with GNN link prediction models to forecast money mule ring formation.

WhatsApp & IVR Shield Bot

Direct deployment of CitizenShield chatbot to WhatsApp Business API and national IVR helplines.

Supporting Research & Industry References

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1. Counterfeit CV (Patil et al. 2022)

Rupee classification CNNs achieve >97% accuracy - proving deep learning is a viable explainable fall back.

2. Audio Spoofing (Li et al. 2023)

Validates vocal jitter & shimmer anomalies as high-precision features for detecting deepfake cloned voices.

3. Graph AML (Linkurious 2024)

Exposes illicit money flows and money mule network clusters that traditional relational checks miss.

4. GIS Hotspots (PolicingInsight 2023)

Emphasizes predictive crime mapping accountability and officer review requirements to avoid bias.

Thank You!

AI for Digital Public Safety Intelligence Platform
Securing Smart Cities, Safeguarding Citizens.

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